



The Hitchhiker's Guide to European Portuguese LLMs

Duarte O.Carmo

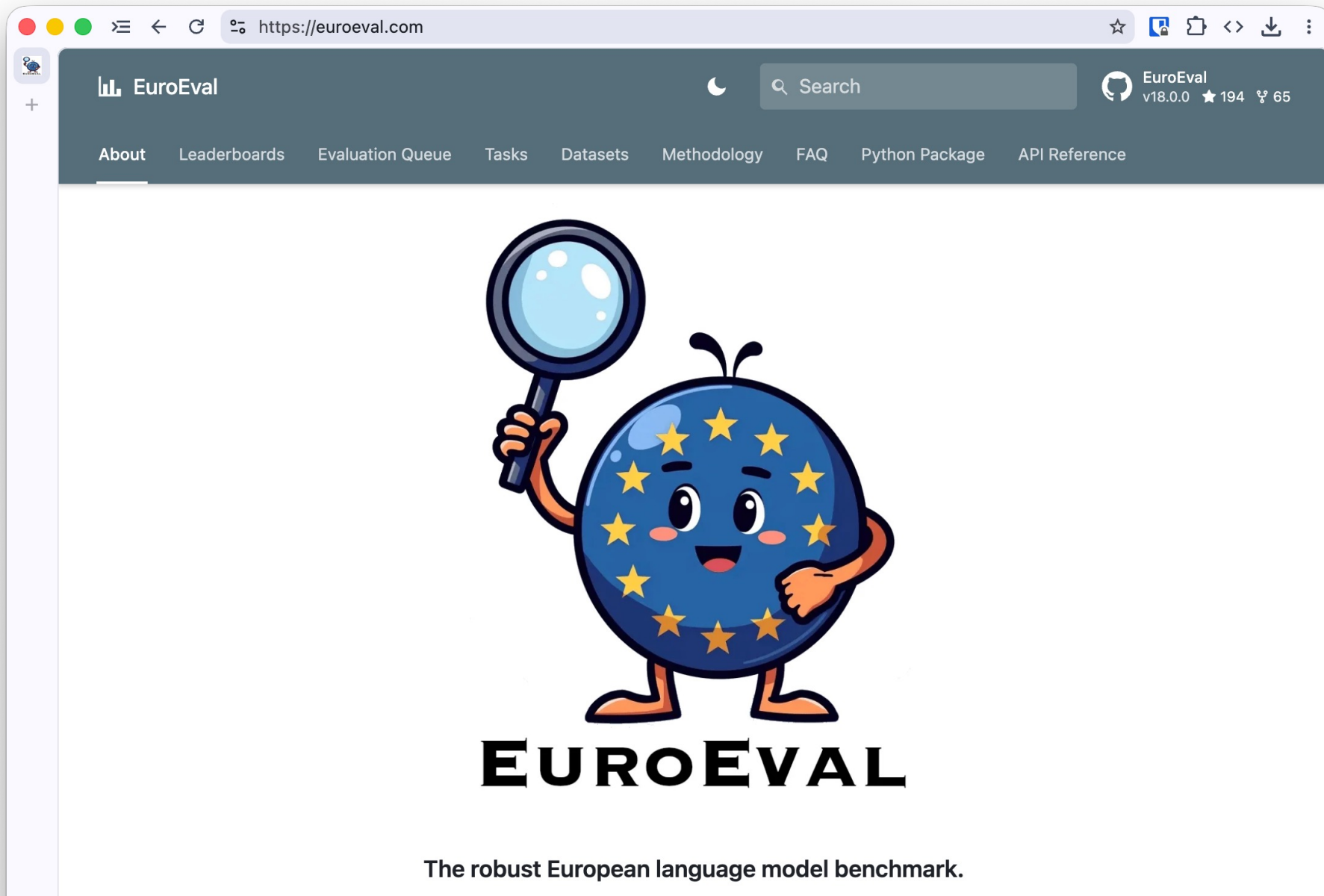
AI/ML Consultant & Researcher @ DOC Tech. | duarteocarmo.com

LISBON AI





What is a good European Portuguese LLM?



**Some months passed, but all I heard
was silence**

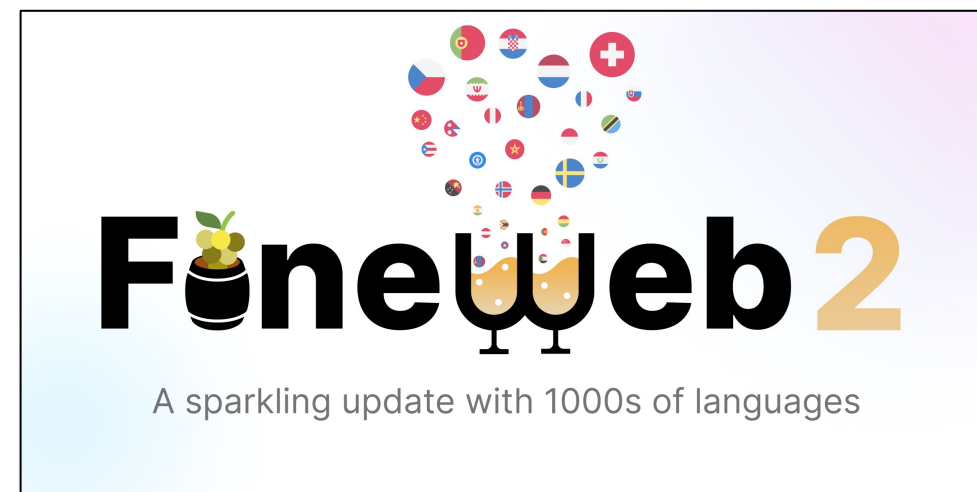
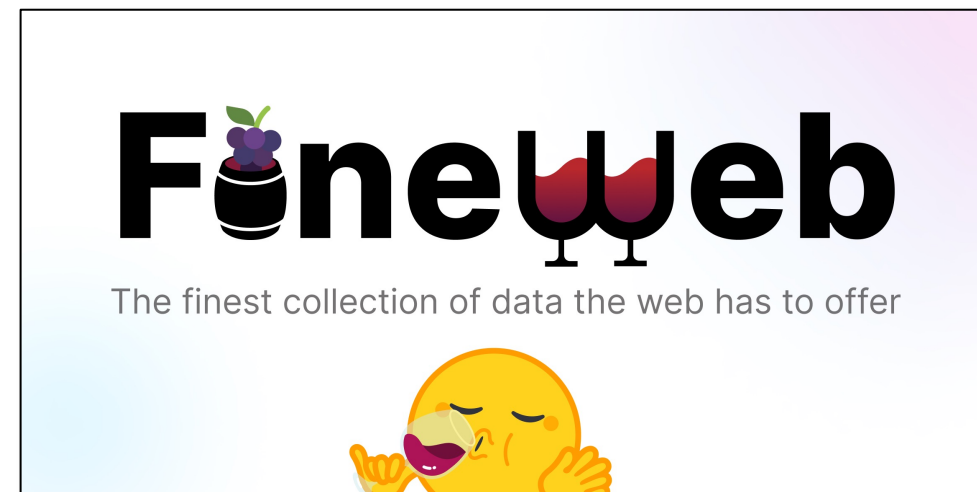
If we're gonna do this, we need **DATA**

FineWeb

55TB of text, **English only**

FineWeb2

20TB of text, but now multilingual 



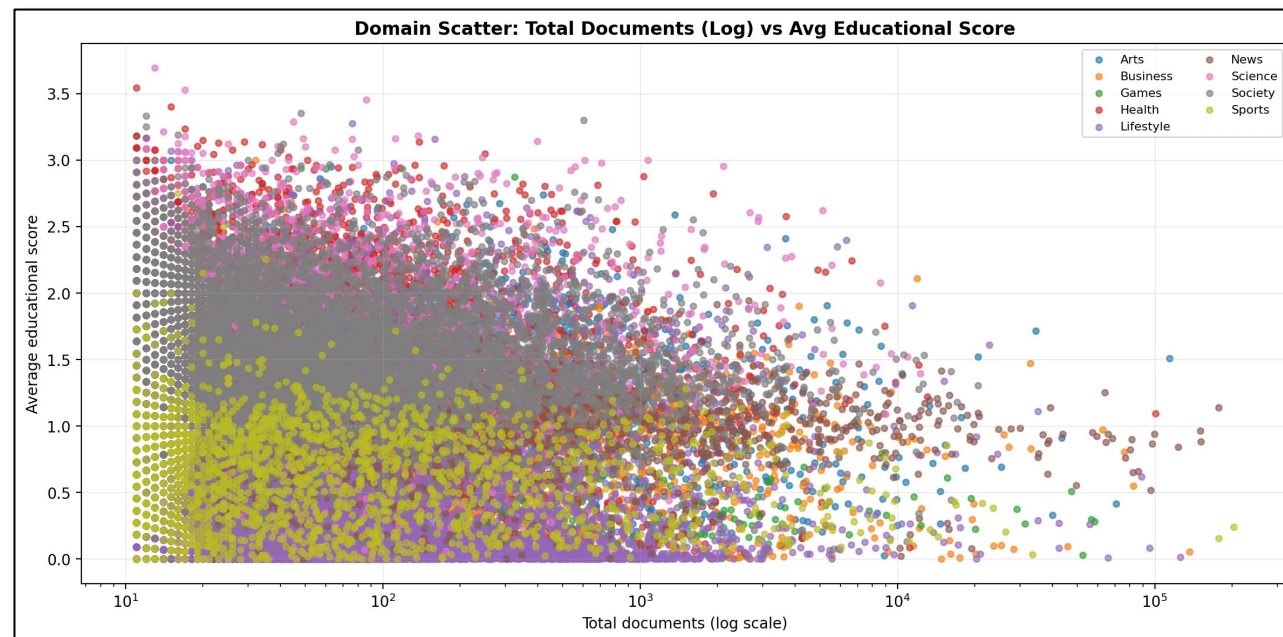
From FineWeb2, I built Bagaço

Extract Portugal from FineWeb2

Take all the .pt domains from FineWeb2

Educational score + Content Category

Result: 9.7B tokens of Portuguese data



**But that approach is stupid,
we can do better – right?**

Enhancing Portuguese Variety Identification with Cross-Domain Approaches

Hugo Sousa^{1,3*}, Rúben Almeida^{2,3,4*}, Purificação Silvano^{5,6},
Inês Cantante^{5,6}, Ricardo Campos^{3, 7,8}, Alípio Jorge^{1,3}

¹Faculty of Sciences, University of Porto, Porto, Portugal

²Faculty of Engineering, University of Porto, Porto, Portugal

³INESC TEC, Porto, Portugal

⁴Innovation Point - dst group, Braga, Portugal

⁵Faculty of Arts and Humanities, University of Porto, Porto, Portugal

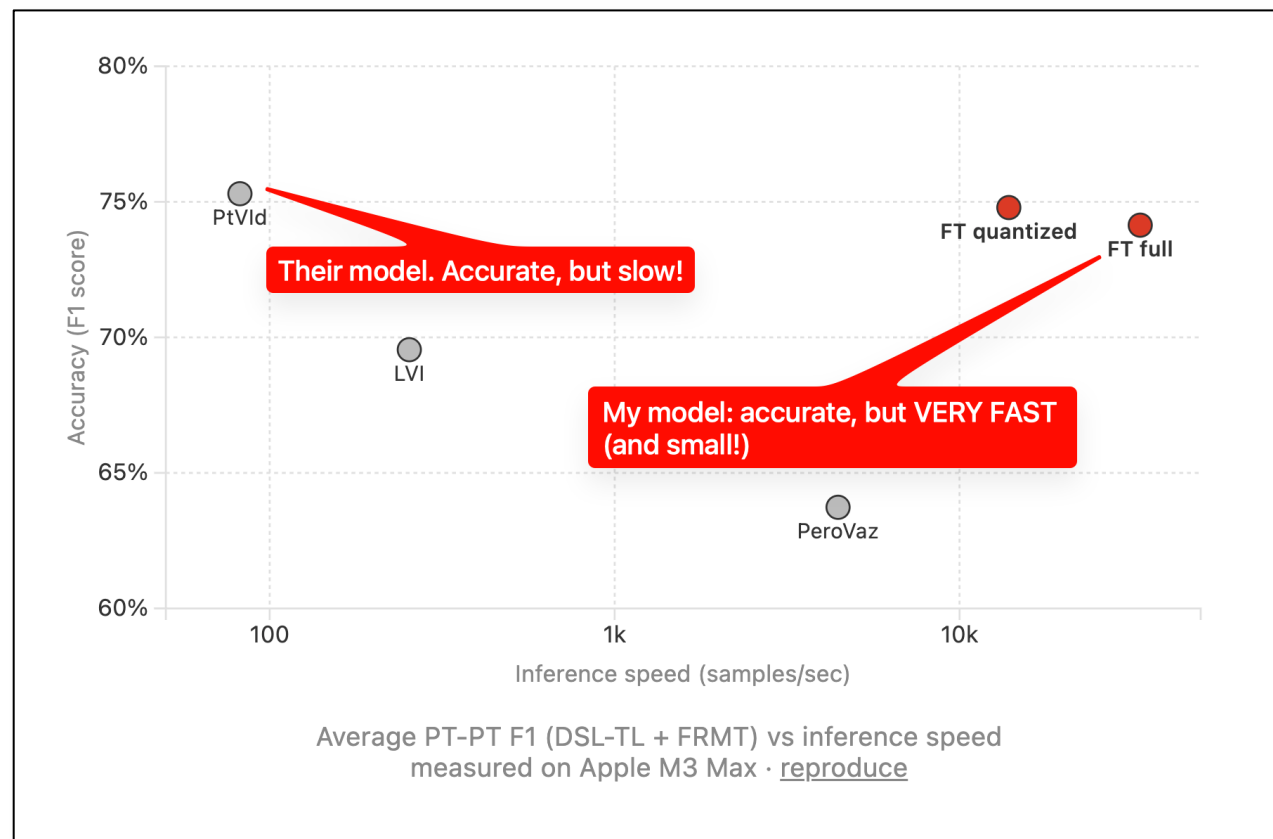
⁶Centre of Linguistics, University of Porto, Porto, Portugal

⁷Department of Informatics, University of Beira Interior, Covilhã, Portugal

⁸Ci2 - Smart Cities Research Center, Tomar, Portugal

hugo.o.sousa@inesctec.pt, ruben.f.almeida@inesctec.pt

A bitter lesson



With that, we can build **Bagaço v2**

Run a pt classifier on top of FineWeb2

Educational score + Content Category

Result: 21.3B tokens of Portuguese data

+119% of data!

Top 10 domains in Bagaço v2 by average educational score (min 500 docs)		
domain	doc_count	avg_educational_score
pt-pt.khanacademy.org	1.3K	3.023
museubiodiversidade.uevora.pt	568	3.002
academia.cienciaviva.pt	515	2.926
casadasciencias.org	2.2K	2.918
avesdeportugal.info	841	2.905
mozaweb.com	924	2.895
educamais.com	850	2.875
alimentacaomedicinal.com	991	2.845
nagwa.com	673	2.841
dicionariosaude.com	1.1K	2.834

**Great – now we can actually build
some models?**



AMALIA Technical Report: A Fully Open Source Large Language Model for European Portuguese

Afonso Simplicio^{1,2}, Gonçalo Vinagre^{1,2}, Miguel Moura Ramos^{3,4}, Diogo Tavares^{1,2}, Rafael Ferreira^{1,2},
Giuseppe Attanasio³, Duarte M. Alves^{3,4}, Inês Calvo¹, Inês Vieira¹, Rui Guerra¹, James Furtado^{1,2},
Beatriz Canaverde^{3,4}, Iago Paulo^{1,2}, Vasco Ramos^{1,2}, Diogo Glória-Silva^{1,2}, Miguel Faria³, Marcos Treviso^{3,4},
Daniel Gomes⁵, Pedro Gomes⁵, David Semedo^{1,2}, André Martins^{3,4}, João Magalhães^{1,2}

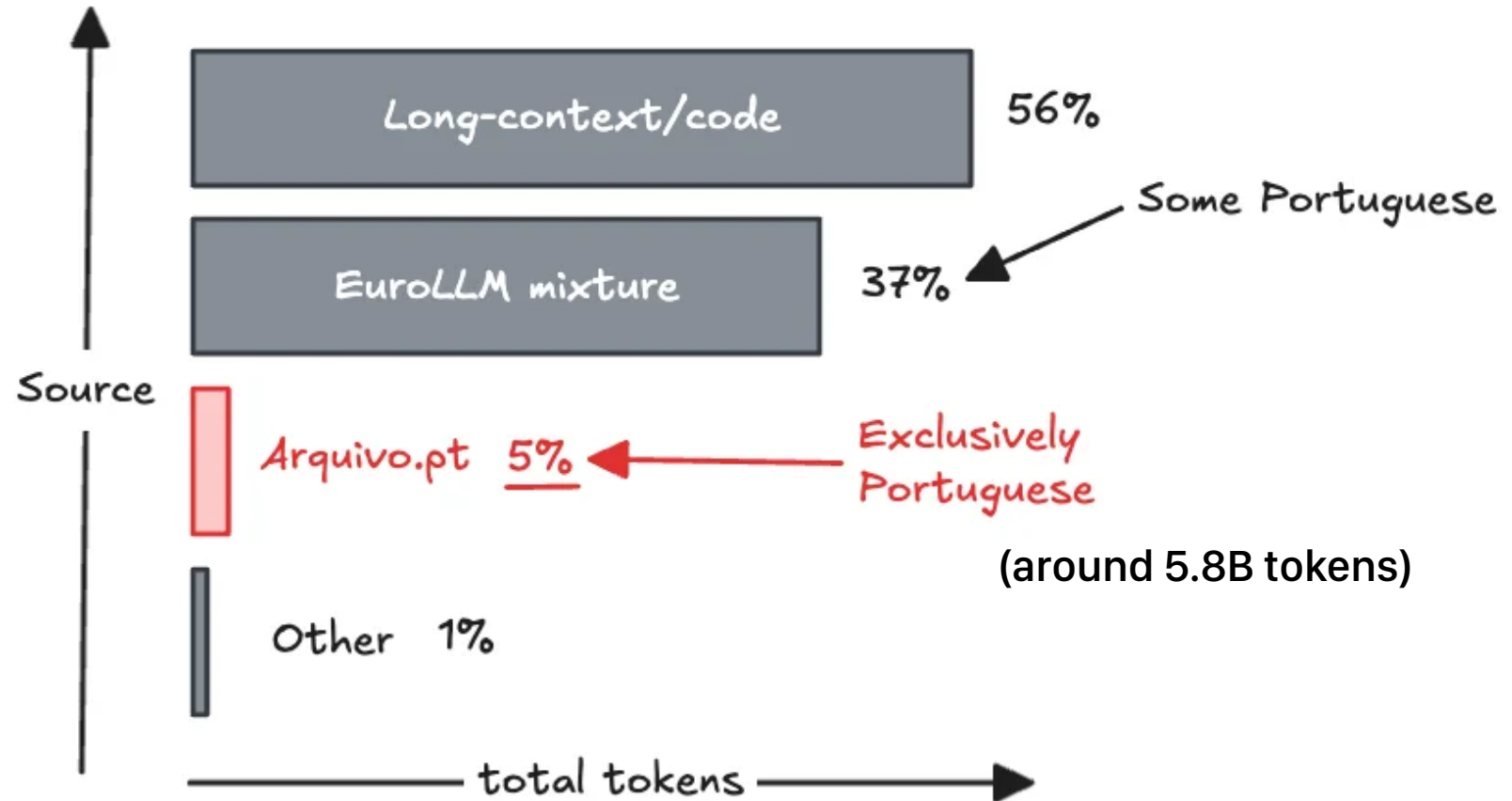
¹NOVA School of Science and Technology, ²NOVA LINCS, ³Instituto de Telecomunicações,

⁴Instituto Superior Técnico, Universidade de Lisboa, ⁵Fundação para a Ciência e Tecnologia

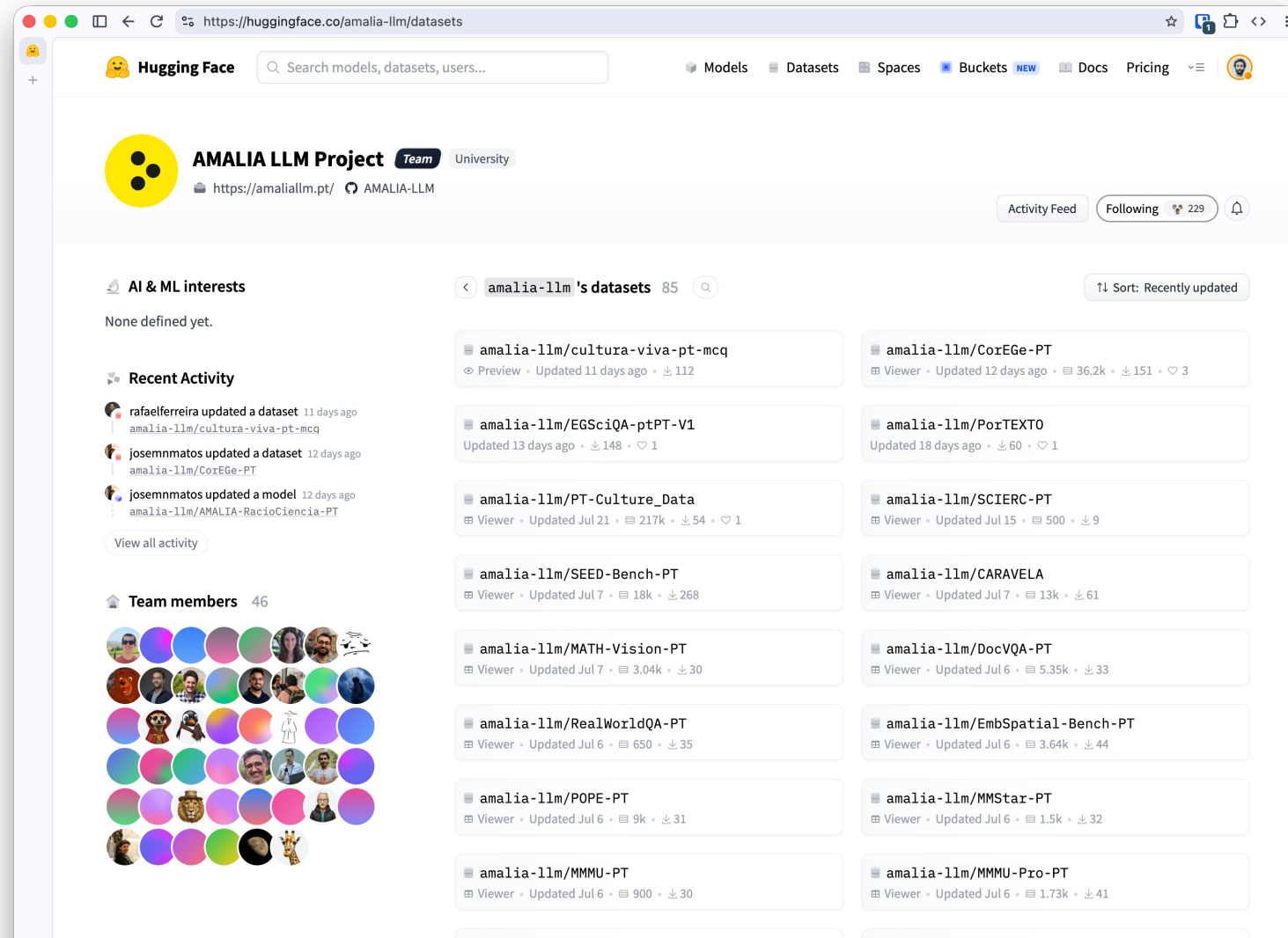
Correspondence: {am.simplicio, gv.martins}@campus.fct.unl.pt

Everyone is thinking about the models...

AMÁLIA continued pretraining data distribution



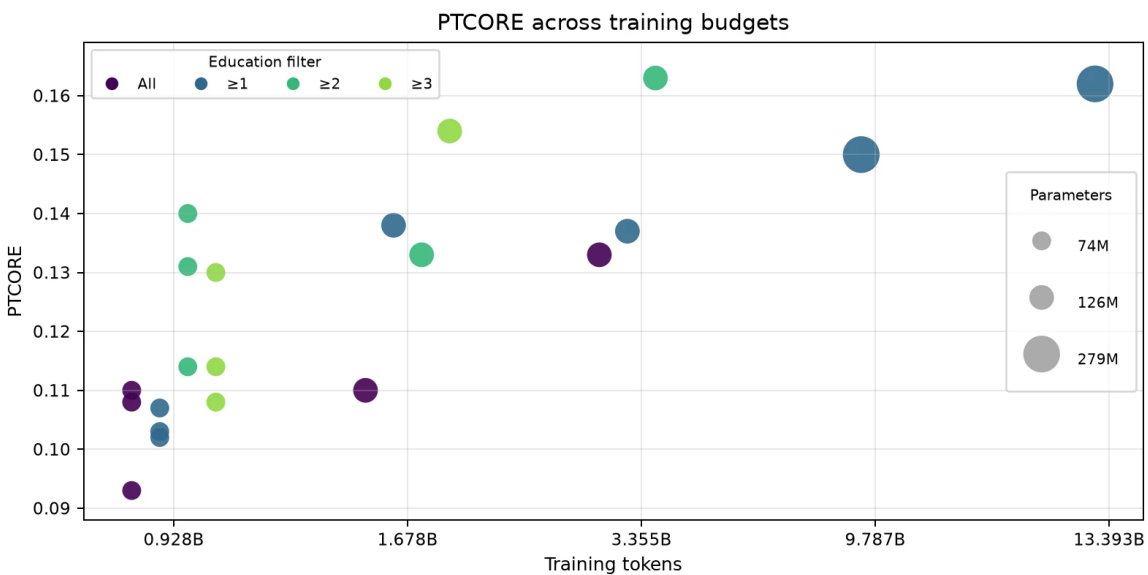
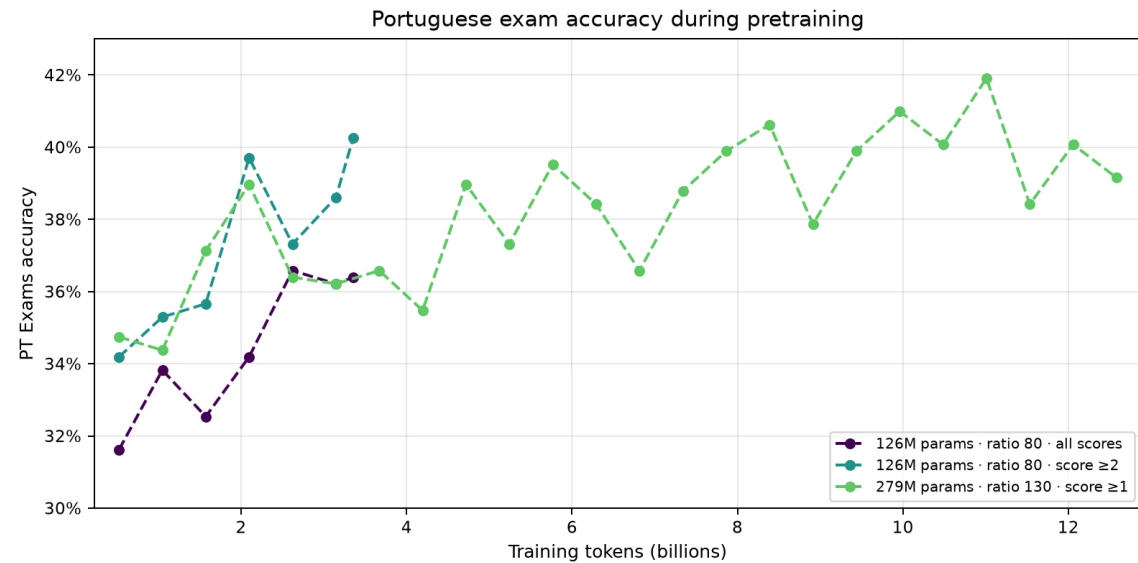
That they forgot about the evals



The screenshot shows the Hugging Face profile for the AMALIA LLM Project. The page displays a list of datasets created by the project, including:

- amalia-11m/cultura-viva-pt-mcq
- amalia-11m/EGSciQA-ptPT-V1
- amalia-11m/PT-Culture_Data
- amalia-11m/SEED-Bench-PT
- amalia-11m/MATH-Vision-PT
- amalia-11m/RealWorldQA-PT
- amalia-11m/POPE-PT
- amalia-11m/MMMU-PT
- amalia-11m/CorEGe-PT
- amalia-11m/PosTEXT0
- amalia-11m/SCIERC-PT
- amalia-11m/CARAVELA
- amalia-11m/DocVQA-PT
- amalia-11m/EmbSpatial-Bench-PT
- amalia-11m/MMStar-PT
- amalia-11m/MMMU-Pro-PT

The page also shows recent activity and team members.



Measuring performance as we pre-train with **PTCORE**

Task	What it measures	Examples
SST2-PT	Sentiment classification in Portuguese	2,048
ALBA-MCQ	European Portuguese linguistics, language variety, and wordplay	240
CulturaVivaPT	Portuguese culture, places, history, and personalities	1,000
PT Exams	History and Geography questions from Portuguese national exams	544
SAUDADE	Temporal reasoning about Portuguese events	8,573
OpenBookQA-PT	Commonsense and elementary science questions translated into Portuguese	500

A problem of data quality (or educational value)

D6 educational-score ablation								
Centered scores × 100 · bold marks the winner within each seed and across means								
Filter	Seed	SST2-PT	ALBA	Cultura	PT Exams	SAUDADE	OpenBookQA	PTCORE
All	42	29.1	16.9	1.9	13.0	-2.0	7.2	11.0
	1337	24.7	21.9	2.0	11.0	-3.4	8.5	10.8
	2026	18.8	16.2	6.0	11.0	-2.6	6.1	9.3
	Mean ± SD	24.2 ± 5.1	18.3 ± 3.1	3.3 ± 2.3	11.7 ± 1.1	-2.7 ± 0.7	7.3 ± 1.2	10.4 ± 0.9
≥1	42	18.7	18.1	3.9	15.0	-2.1	8.0	10.2
	1337	23.4	18.1	5.2	12.3	-1.0	6.4	10.7
	2026	20.9	18.1	2.9	15.0	-1.8	6.7	10.3
	Mean ± SD	21.0 ± 2.4	18.1 ± 0.0	4.0 ± 1.1	14.1 ± 1.6	-1.7 ± 0.6	7.0 ± 0.9	10.4 ± 0.3
≥2	42	27.0	24.4	4.8	15.4	3.3	9.1	14.0
	1337	26.0	21.2	5.1	15.4	1.9	9.1	13.1
	2026	16.5	24.4	4.9	12.3	2.7	7.5	11.4
	Mean ± SD	23.1 ± 5.8	23.3 ± 1.8	4.9 ± 0.1	14.4 ± 1.8	2.6 ± 0.7	8.5 ± 0.9	12.8 ± 1.3
≥3	42	16.7	20.0	5.2	15.0	4.3	7.2	11.4
	1337	11.4	18.8	7.2	13.7	4.7	8.8	10.8
	2026	22.4	22.5	5.1	15.2	4.9	8.0	13.0
	Mean ± SD	16.8 ± 5.5	20.4 ± 1.9	5.8 ± 1.2	14.6 ± 0.8	4.6 ± 0.3	8.0 ± 0.8	11.7 ± 1.2

Educational score	Count	Percentage
0	14,926,917	45.15%
1	11,357,279	34.35%
2	4,536,413	13.72%
3	2,001,114	6.05%
4	237,073	0.72%

Bagaço v3 the latest version

Adds FinePDFs and FineWiki to version 2

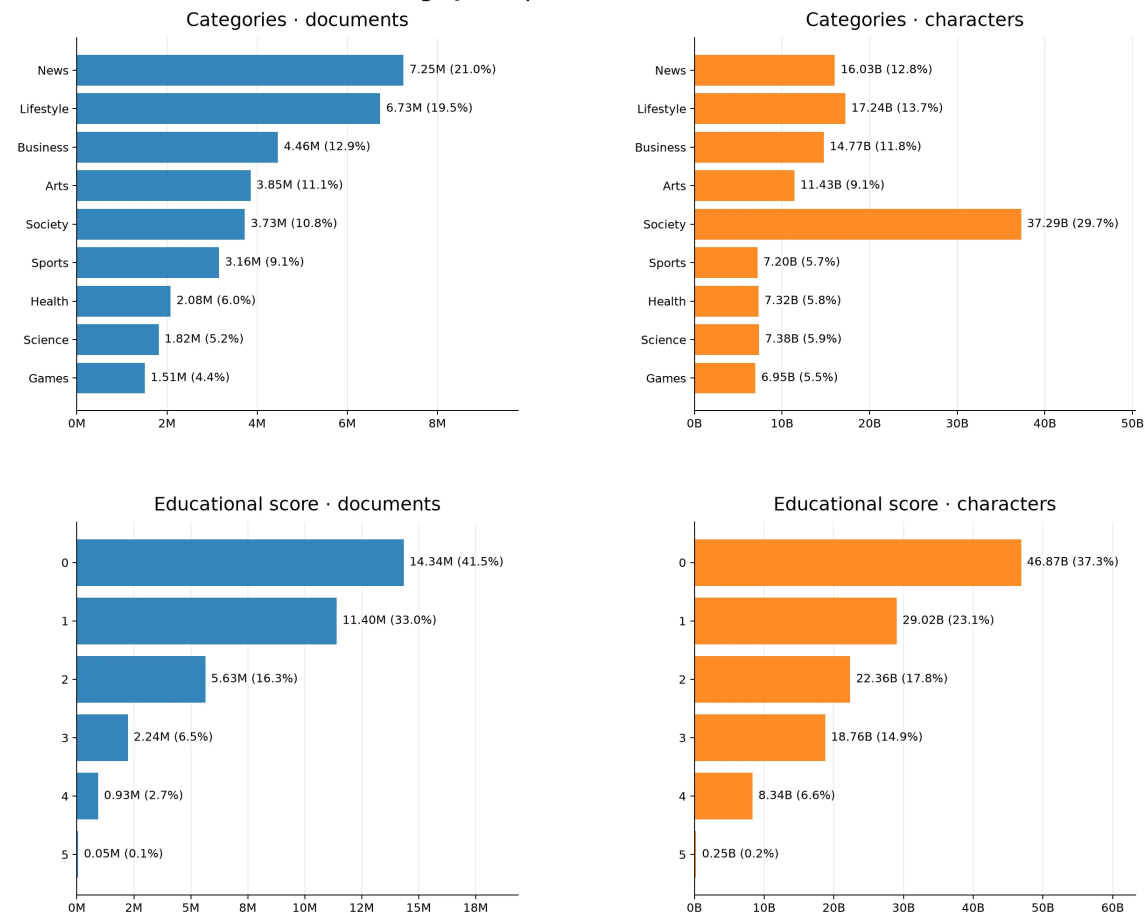
Educational score + Content Category

Result: 29B tokens of Portuguese data

+37% of data from version 2!

Version	Documents	Estimated tokens	Increase	95% CI
Bagaço v1	16,415,860	9.73B	Baseline	9.70B–9.77B
Bagaço v2	33,058,796	21.34B	+119.2%	21.24B–21.44B
Bagaço v3	34,582,524	29.3B	+37.3%	28.9B–29.9B

Bagaço v3 predicted labels



What is a good European Portuguese LLM?

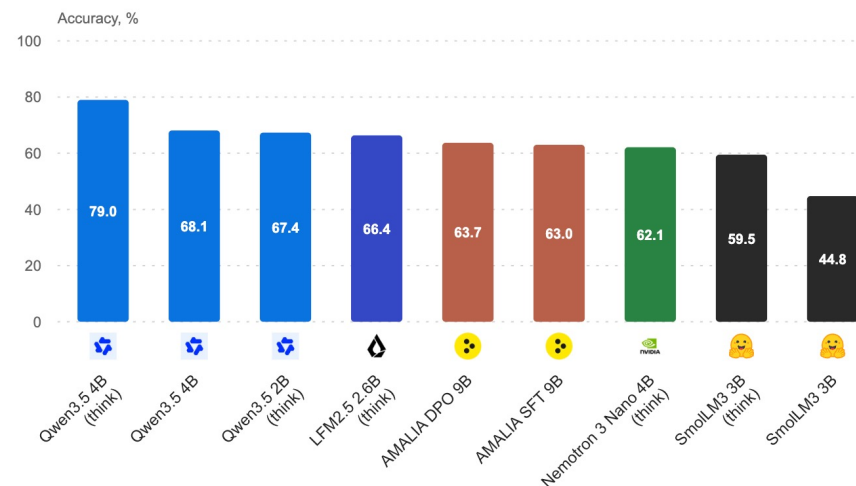
We didn't answer
the question



Portuguese benchmark intelligence

Average of 4 benchmark accuracies · 4,158 questions

All benchmarks (N=4,158) ▾



If you know a thing or two about LLMs, you'll probably call bullshit on this right away. Yes, the other models are smaller, but they're also *reasoning* models (most of them), while AMALIA is *not*. They use something called test-time scaling to improve their performance. This basically means that they let the LLM output more tokens when it's time to answer the question, leading to better performance.

You'll say it's a little unfair to compare AMALIA with these. They come from big labs, from NVIDIA to Alibaba's Qwen team. AMALIA has a smaller team, a smaller budget, and admittedly a narrower goal. I agree. Still, it's interesting to see much smaller models outperform it even on Portuguese-focused tasks.

So what?

More questions than answers

Are we missing data for European Portuguese?

What do we consider a “good” model?

As a business what would make you adopt AMALIA?

Is a smaller reasoning/specialized model the way to go?

Thanks!